ALM - 3 (CO - 2)

1. What is a regression line?

Ans. A regression line is a straight line that best represents the relationship between two variables. It shows the average change in one variable corresponding to a unit change in ~~the~~ another and is used for prediction and trend analysis.

2. What is the range of regression coefficients?

Ans. Regression coefficients can take any real value from negative ~~infi~~ infinity to positive infinity. Unlike correlation coefficients, they are not restricted between -1 and +1. Their values depend on the units and scale of the variables.

3. Can both regression coefficients be greater than 1?

Ans. No, both regression coefficients cannot be greater than 1 simultaneously. The product of two regression coefficients is equal to the square of the correlation, which is always less than or equal to 1.

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4. What happens to regression lines when correlation is perfect?

Ans. When corellation is perfect ($r = +1$ or $r = -1$) both regression lines coincide & become the same straight line. This means every observed value lies exactly on the line and prediction becomes completely accurate.

5. What is the purpose of regression analysis?

Ans. The purpose of regression analysis is to study the relationship between variables and to predict the value of one variable based on another. It helps in forecasting, decision making, and understanding how changes in one variable affect others.

6. When will the regression lines be perpendicular to each other?

Ans. Regression lines are perpendicular to each other when the correlation coefficient is zero ($r = 0$). In this case, the variables are uncorrelated, and the two regression equations intersect at right angles.

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